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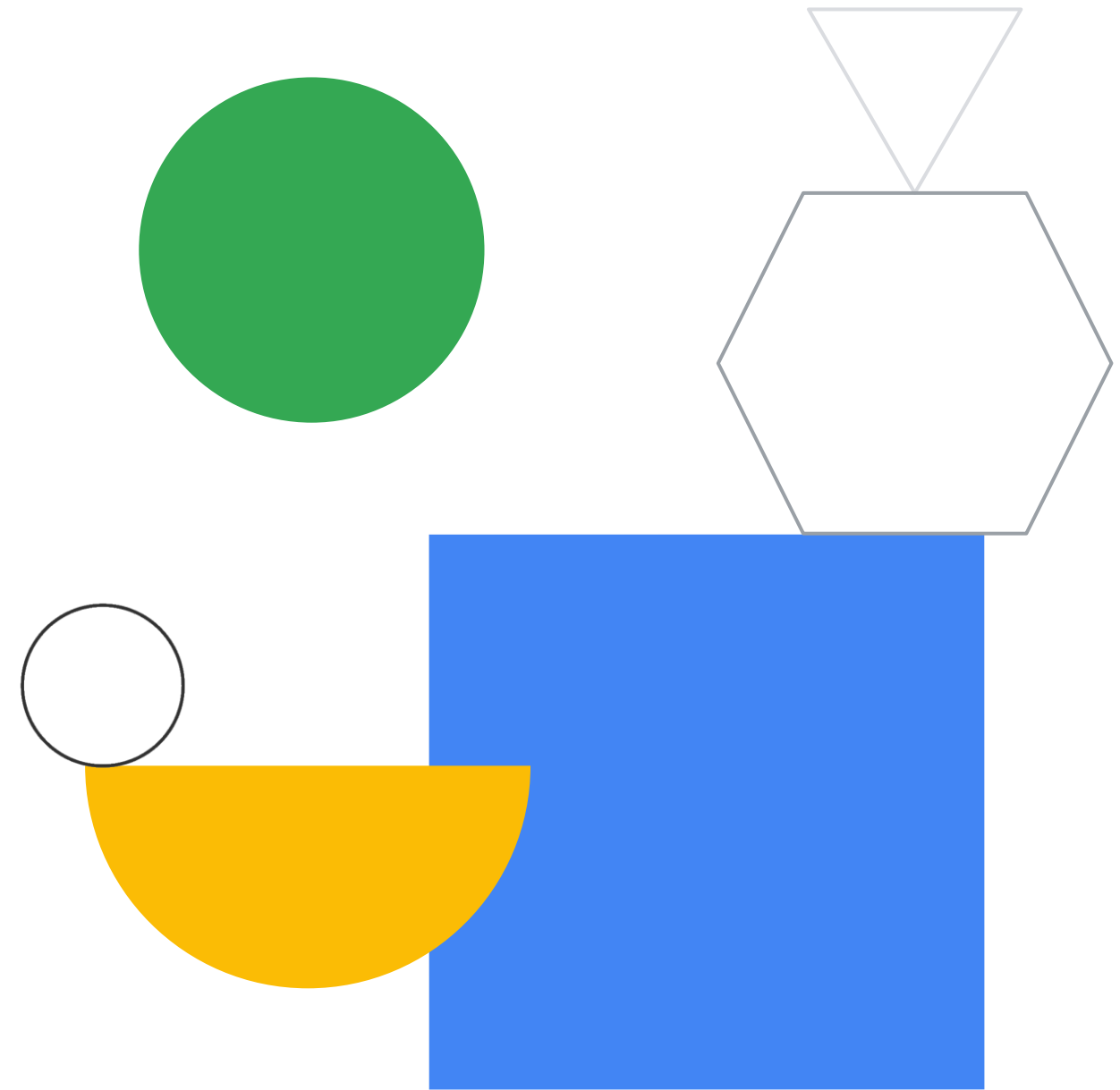
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You have the data, but what are you doing with it?

Module 9 - Big data
Google Cloud Computing Foundations



Course map

01 —

So, what's the cloud anyway?

02 —

Start with a solid platform

03 —

Use Google Cloud to build your apps

04 —

Where do I store this stuff?

05 —

There's an API for that!

06 —

You can't secure the cloud, right?

07 —

It helps to network

08 —

Keeping an eye on things

09 —

You have the data, but what are you doing with it?

10 —

Let machines do the work

Google Cloud skill badges

Objectives

- 01 Explore big data managed services in the cloud.
- 02 Examine using Dataproc to run Hadoop, Spark, and other big data technologies as a managed service in the cloud.
- 03 Learn about building ETL pipelines as a managed service using Dataflow.
- 04 Explore BigQuery as a managed data warehouse and analytics engine.



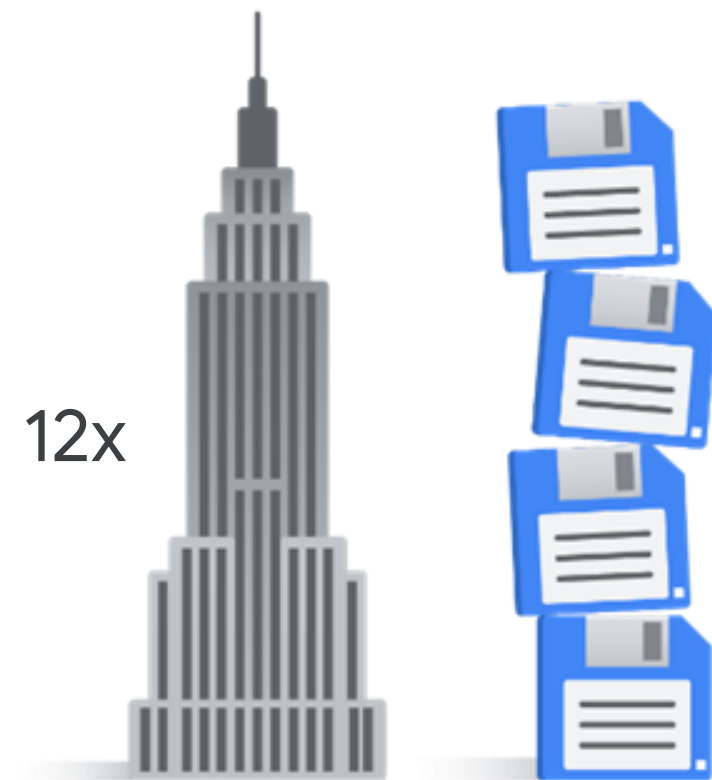
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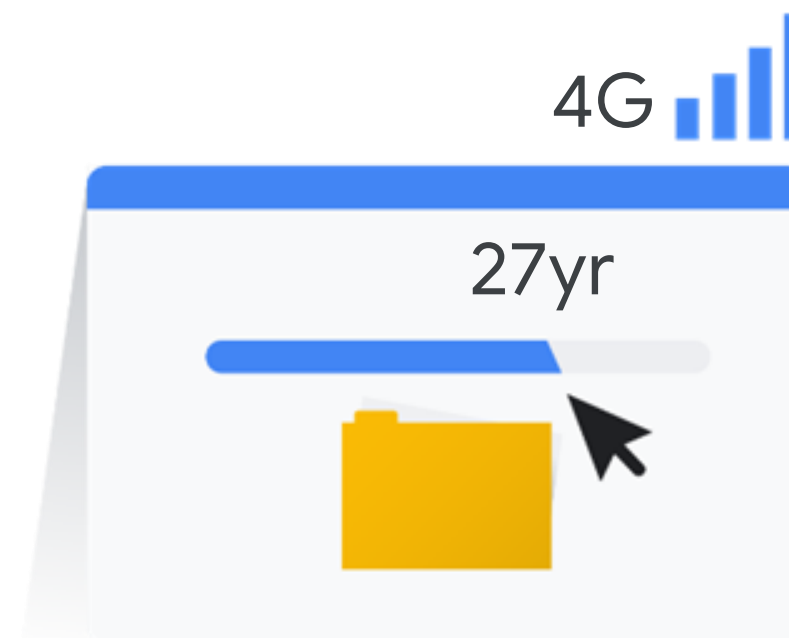
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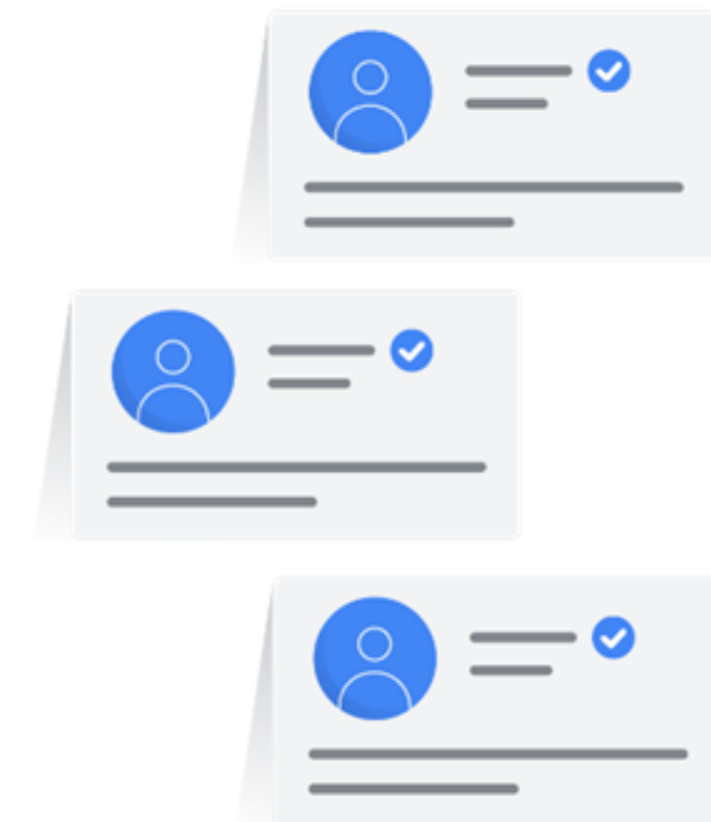
How big is a petabyte of data?



A stack of floppy disks
higher than **12** Empire
State Buildings



27 years to
download over 4G



Every tweet ever
tweeted ... **x 50**

How small is a petabyte of data?

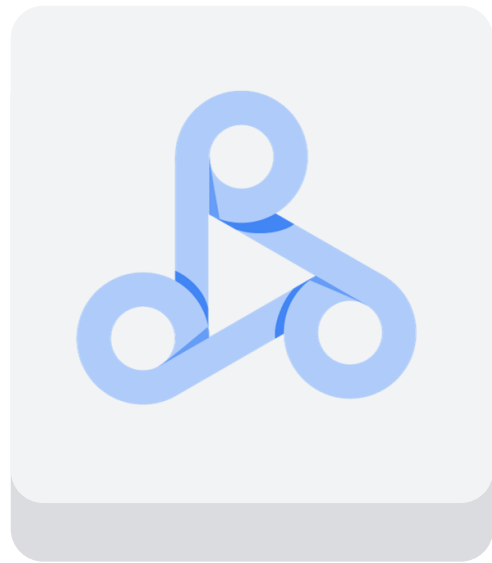


2 micrograms of DNA



1 day's worth of video
uploaded to YouTube

Overview of big data managed services



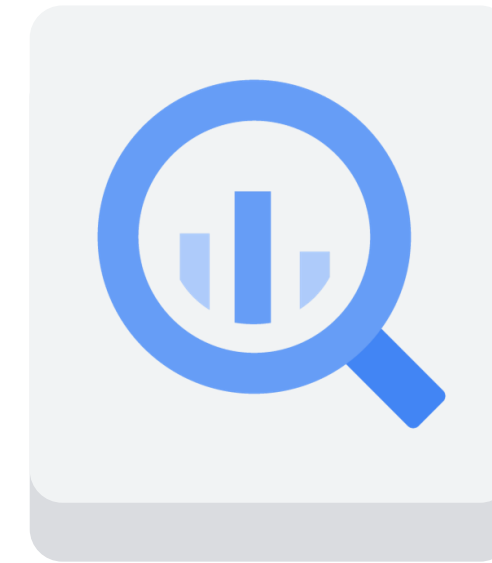
Dataproc

Process big data with
Hadoop/Spark



Dataflow

Analyze streaming
data in real time



BigQuery

Modernize a data
warehouse foundation

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Hadoop and Spark are open source technologies that often are the foundation of big data processing

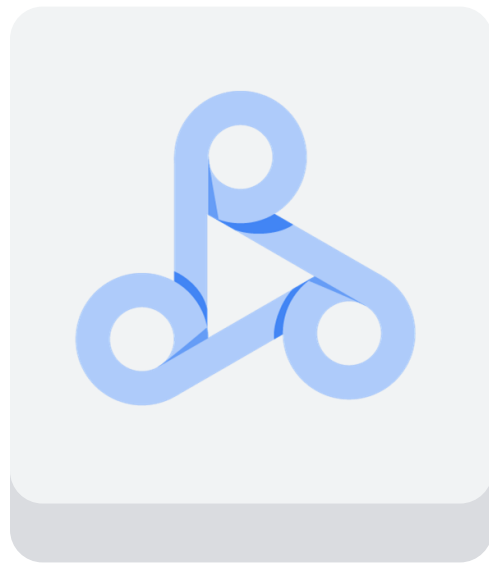
Apache Hadoop



Apache Spark



Dataproc is a managed service for Spark and Hadoop



Dataproc

- ✓ Cost effective
- ✓ Fast and scalable
- ✓ Open source ecosystem
- ✓ Fully managed
- ✓ Image versioning
- ✓ Built-in integration

Dataproc use case 1: Log processing

The need

Large volumes of data from several sources are aggregated each day and loaded into databases so metrics can be gathered.

A dedicated on-premises cluster is used to store and process the logs with MapReduce.

The solution

Cloud Storage provides a low-cost storage option.

An ephemeral Dataproc cluster can be created in less than 2 mins.

Data is processed with existing MapReduce.

The value

Saves money and reduces complexity.

Dataproc use case 2: Ad-hoc data analysis

The need

Analysts are using Spark Shell but are concerned about increase in usage.

Unsure on how to scale their cluster, which is running in standalone mode.

The solution

Creates clusters that scale for speed and mitigate failure.

Can use web interface, Cloud SDK, or native Spark Shell via SSH.

The value

Unlocks the cloud without technical complexity.

Complex computations take seconds, not hours.

Dataproc use case 3: Machine learning

The need

Spark Machine Learning Libraries (MLlib) are used to run classification algorithms on large datasets.

There is a reliance on cloud based machines to install and customize Spark.

The solution

Spark and MLlib can be installed on any Dataproc cluster.

Customizations can be applied to clusters via initialization actions.

Use Cloud Monitoring to monitor workflows.

The value

Resources can focus on data, not cluster creation and management.

Integration with Google Cloud unlocks new Spark features.

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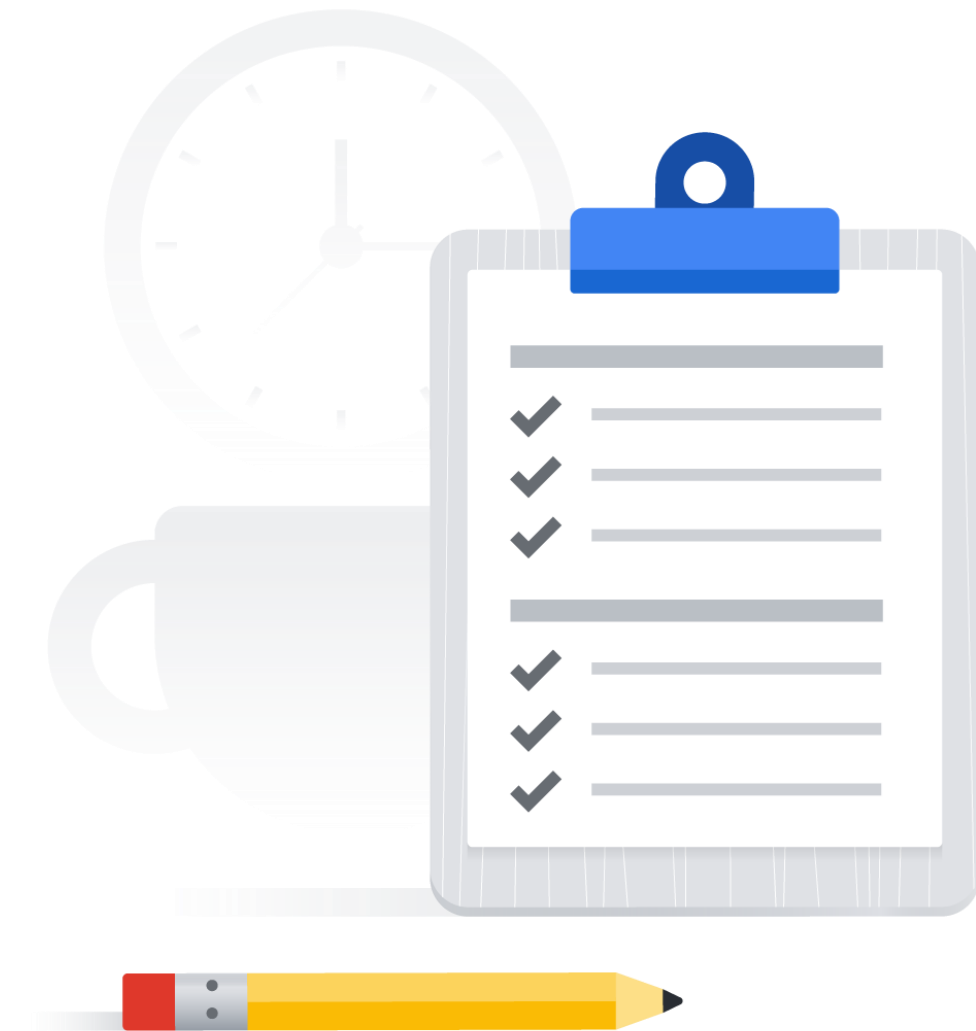
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Lab: Dataproc: Qwik Start - Console

 Hands-on lab

During this lab you'll:

1. Create a cluster.
2. Submit a job.
3. View the job output.



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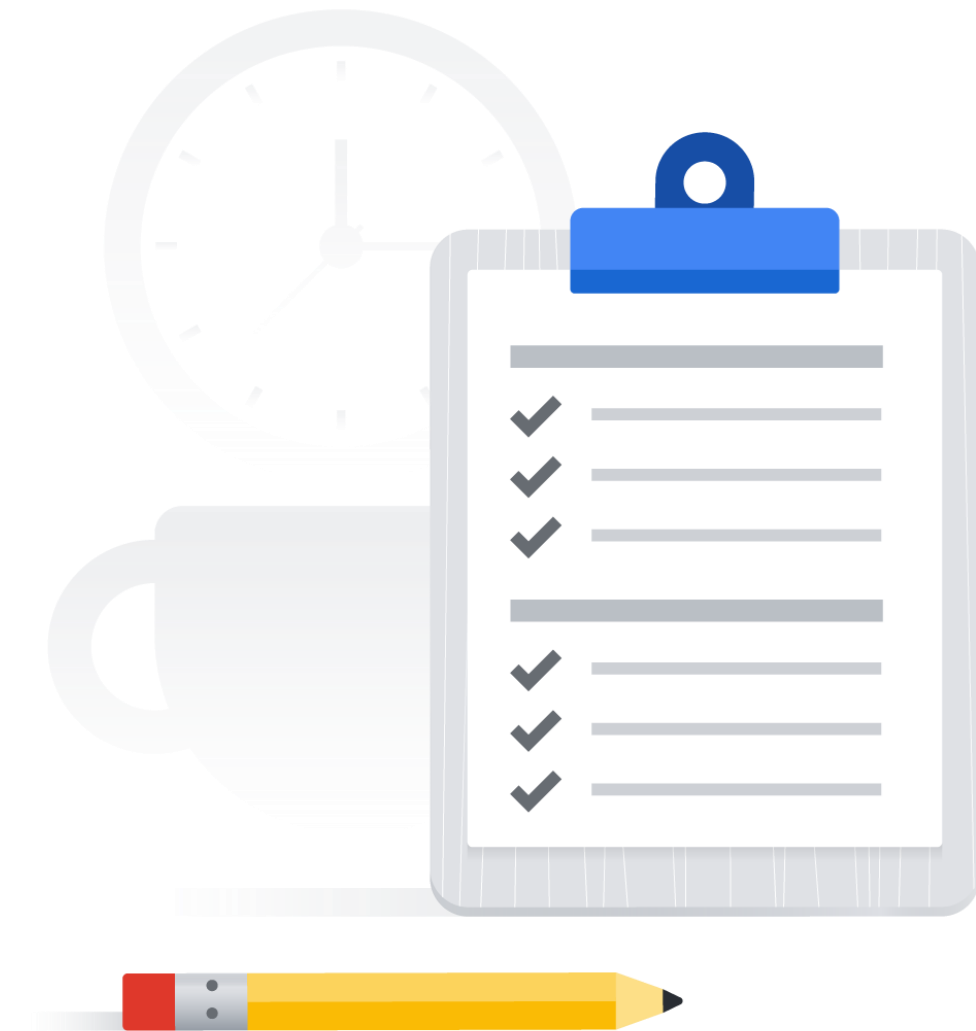
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Lab: Dataproc: Qwik Start - Command Line

 Hands-on lab

During this lab you'll:

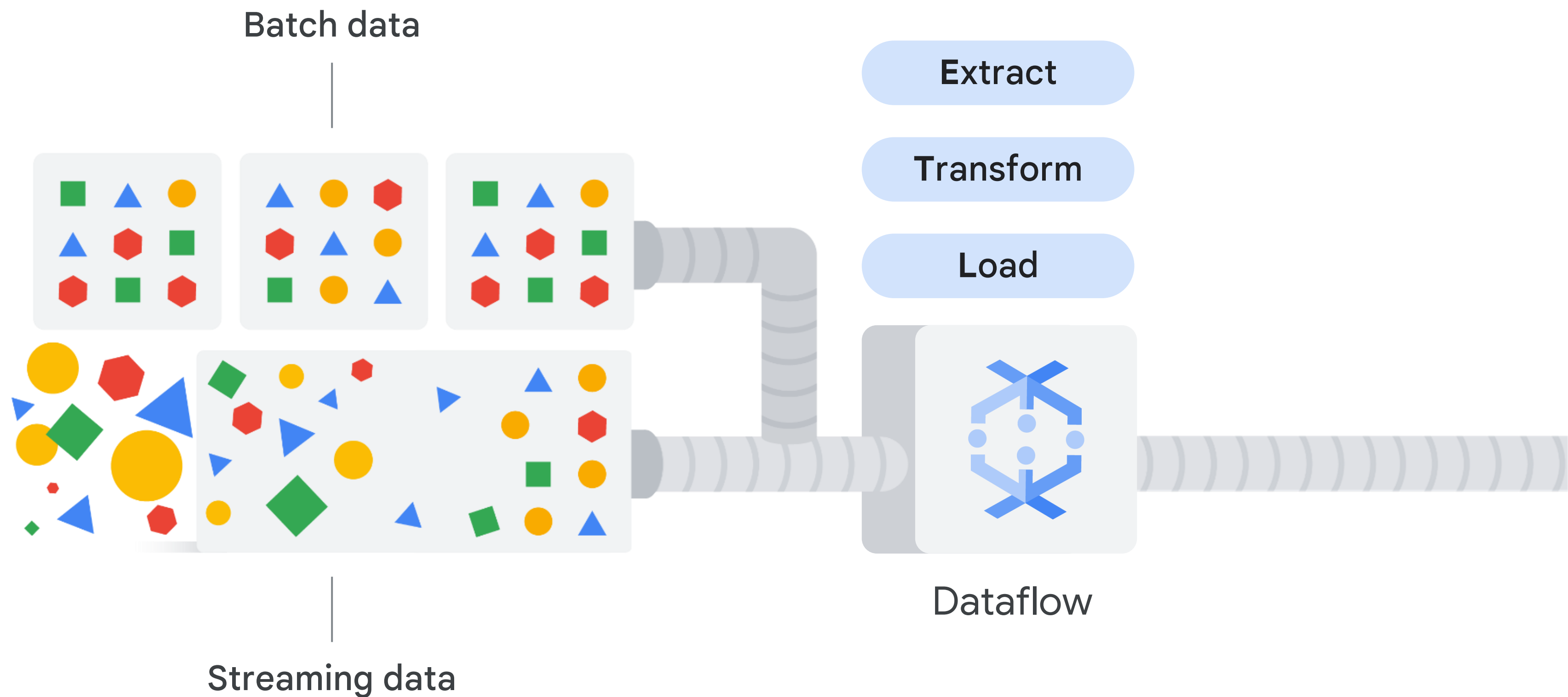
1. Create a cluster.
2. Submit a job.
3. Update a cluster.



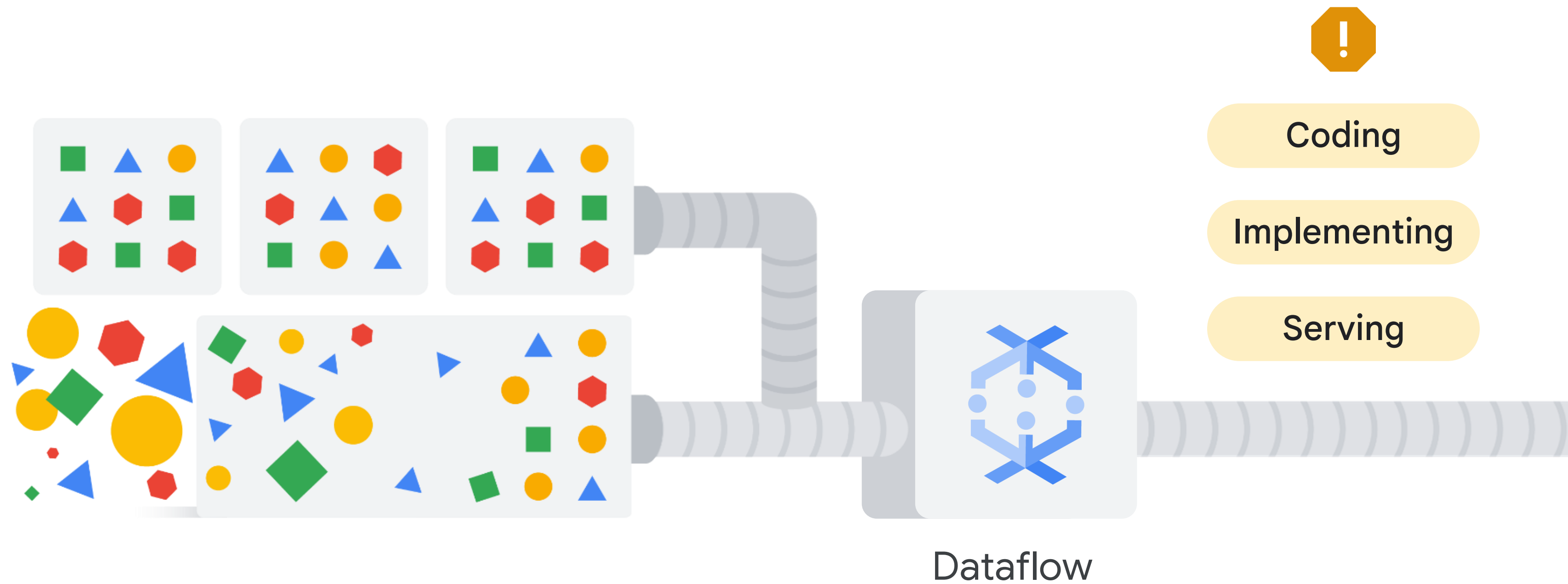
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Dataflow creates a pipeline to process data



Pipeline challenges



Questions to consider during pipeline design



01

Will the pipeline code be compatible with both batch and streaming data, or will it need to be refactored?

02

Will the pipeline code SDK have all the transformations, mid-flight aggregations, and windowing?

03

Will the pipeline code SDK be able to handle late data?

04

Are there existing templates or solutions that should be referenced?



Dataflow



Fully automates operational tasks like resource management and performance optimization.



Provides built-in support for fault-tolerant execution.



Through the Google Cloud console, Dataflow provides near-real time statistics such as pipeline throughput and lag, and consolidated worker log inspection.



Integrates with Cloud Storage, Pub/Sub, Datastore, Cloud Bigtable, and BigQuery for seamless data processing.

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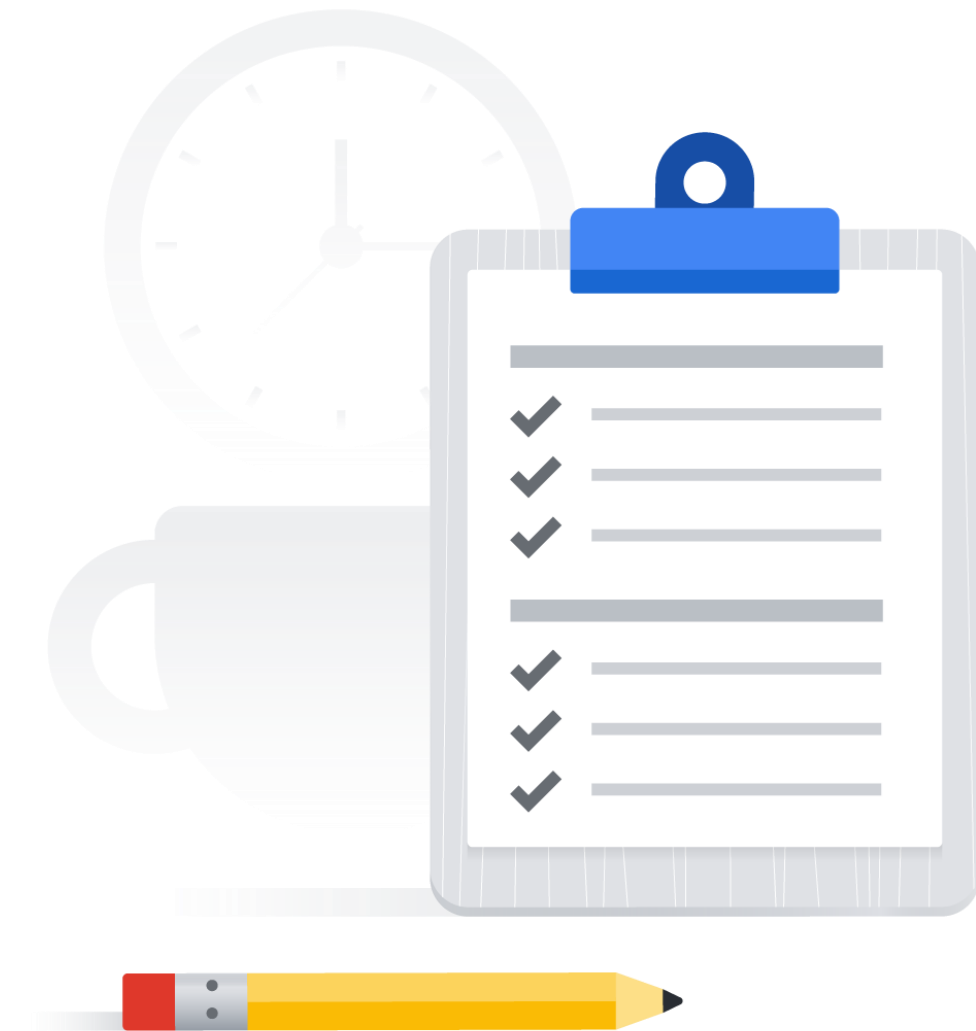
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Lab: Dataflow: Qwik Start - Templates

 Hands-on lab

During this lab you'll:

1. Create a BigQuery dataset and table using Cloud Shell and/or the Google Cloud console.
2. Run the pipeline.
3. Submit a query.



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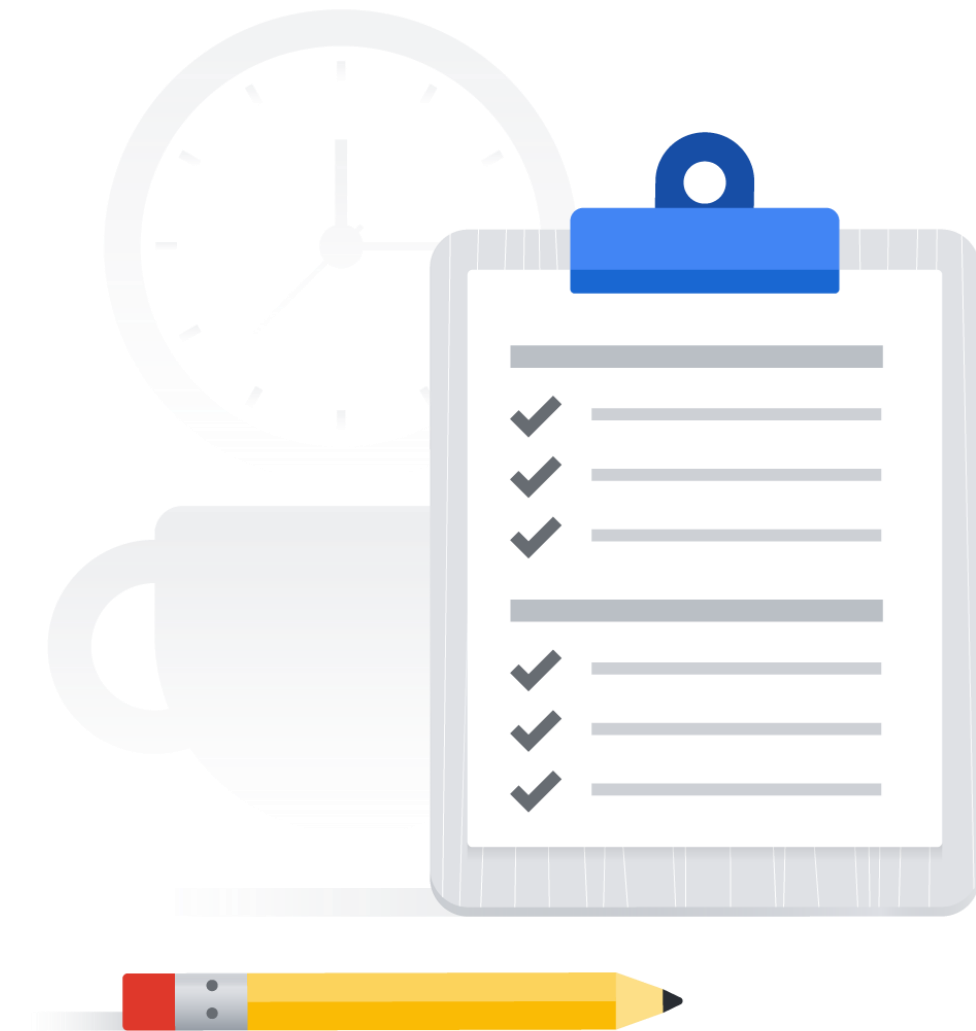
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Lab: Dataflow: Qwik Start - Python

 Hands-on lab

During this lab you'll:

1. Set up a Python development environment.
2. Get the Dataflow SDK for Python.
3. Run an example pipeline using the Google Cloud console.



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BigQuery is a fully-managed data warehouse

Fully managed means that BigQuery takes care of the underlying infrastructure



Terabytes and petabytes of data gathered from a wide range of sources



BigQuery

01

Storage *plus* analytics

A place to store petabytes of data.
(Petabyte = 11,000 4k movies)

A place to analyze data.
(Machine learning, geospatial analysis, and business intelligence)

02

Serverless

Use SQL queries to answer questions.

03

Flexible pay-as-you-go pricing model

Pay for what your query processes, or use a flat-rate option.

04

Data encryption at rest by default

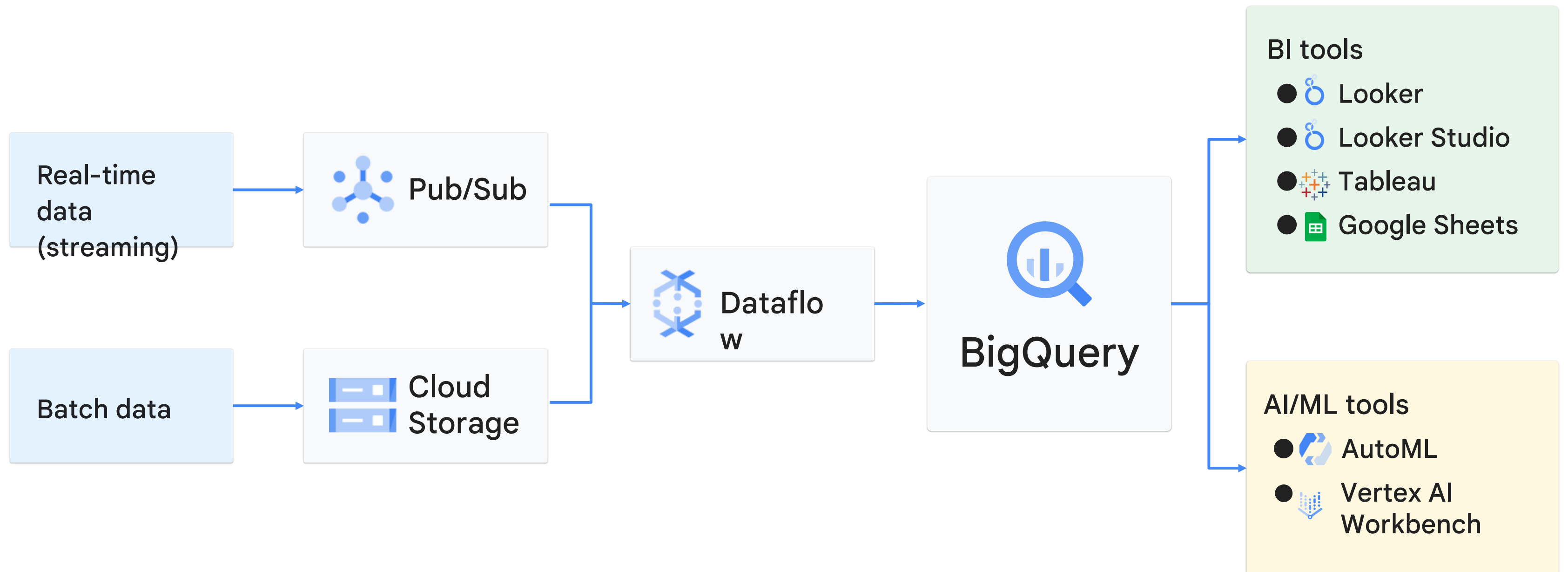
Encrypt data stored on a disk, including SSDs and backup media.

05

Built-in machine learning

Write ML models directly in BigQuery using SQL.

Google data warehouse solution architecture



BigQuery can ingest datasets from various sources

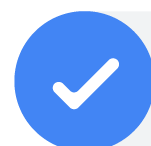
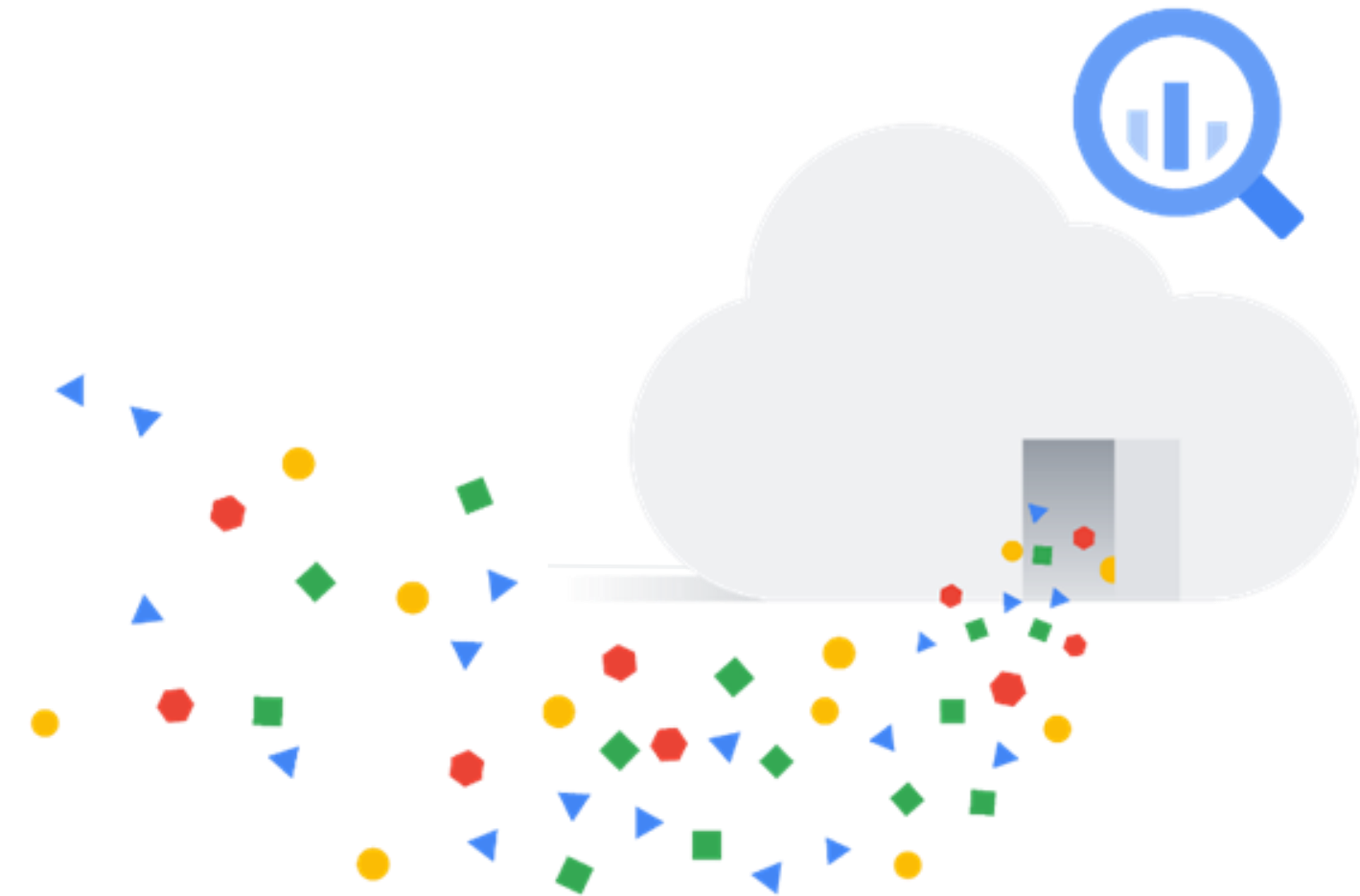
Internal data: Data saved in BigQuery

External data:

- Google Cloud storage (e.g. Cloud Storage)
- Google Cloud database (e.g. Spanner or Cloud SQL)

Multi-cloud data: such as AWS and Azure

Public datasets: Available on the Google Cloud Marketplace



Replicated



Backed up



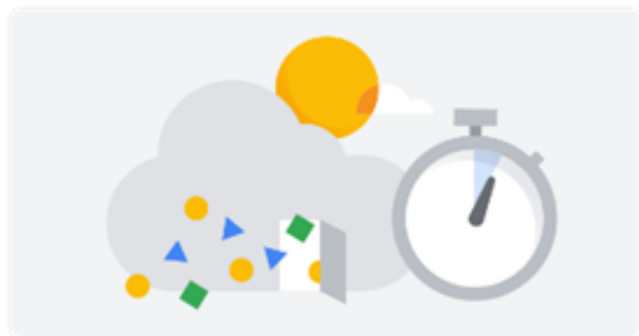
Autoscaled

Patterns to load data into BigQuery

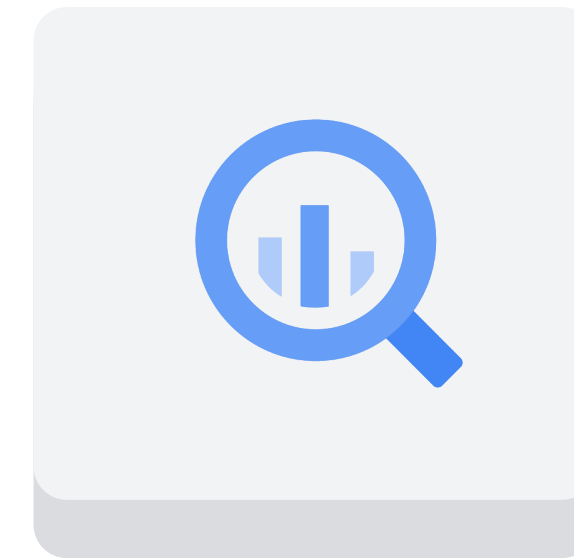
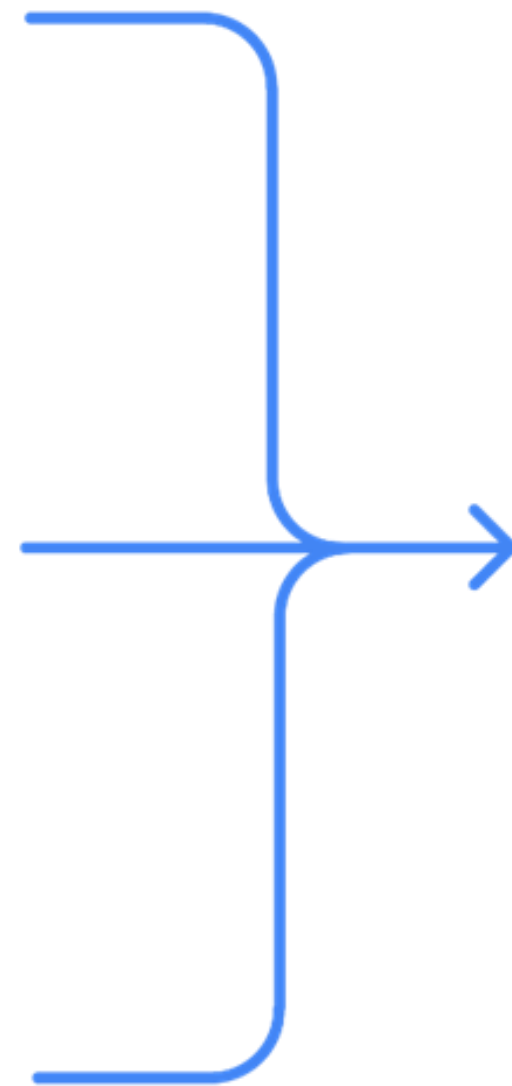
Batch load



Streaming

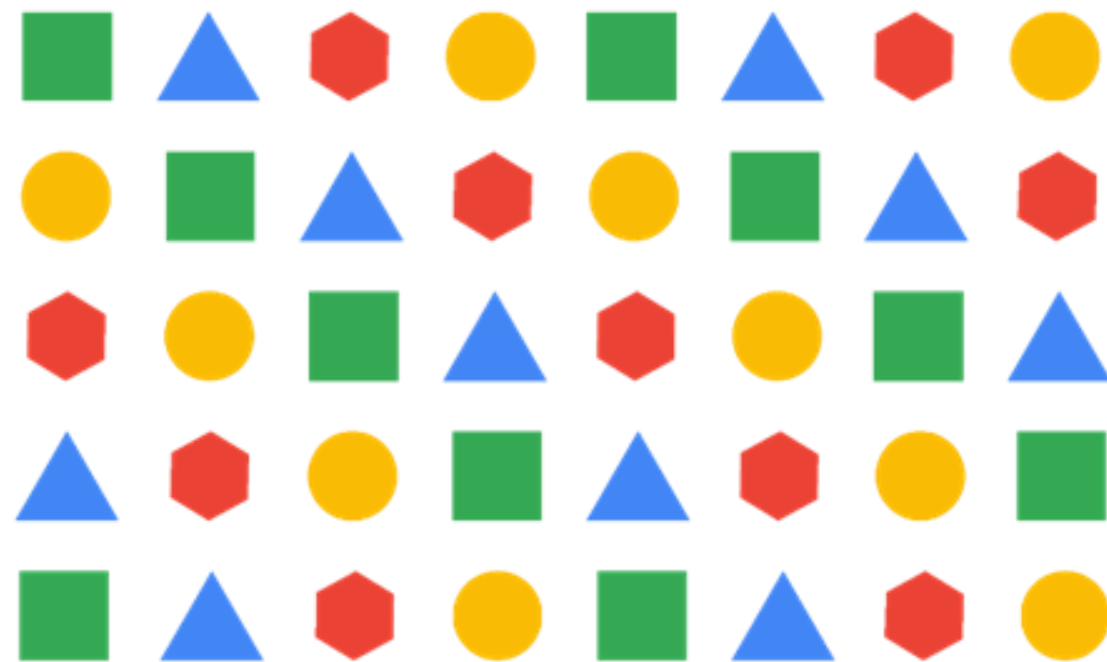


Generated data



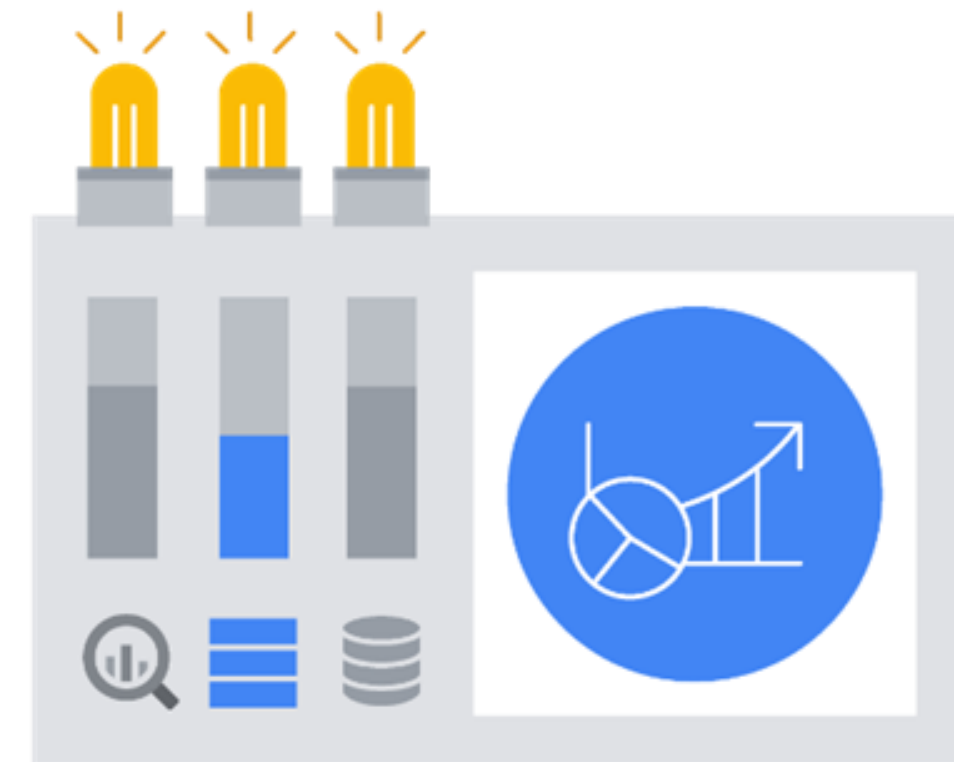
BigQuery

BigQuery can run analytic queries over large datasets



Query terabytes in seconds

Query petabytes in minutes



Analyze data

BigQuery ML



- ✓ Capabilities to build ML models
- ✓ Empowers data analysts
- ✓ No need to export data
- ✓ No reliance on data scientists
- ✓ Accessed in a number of ways

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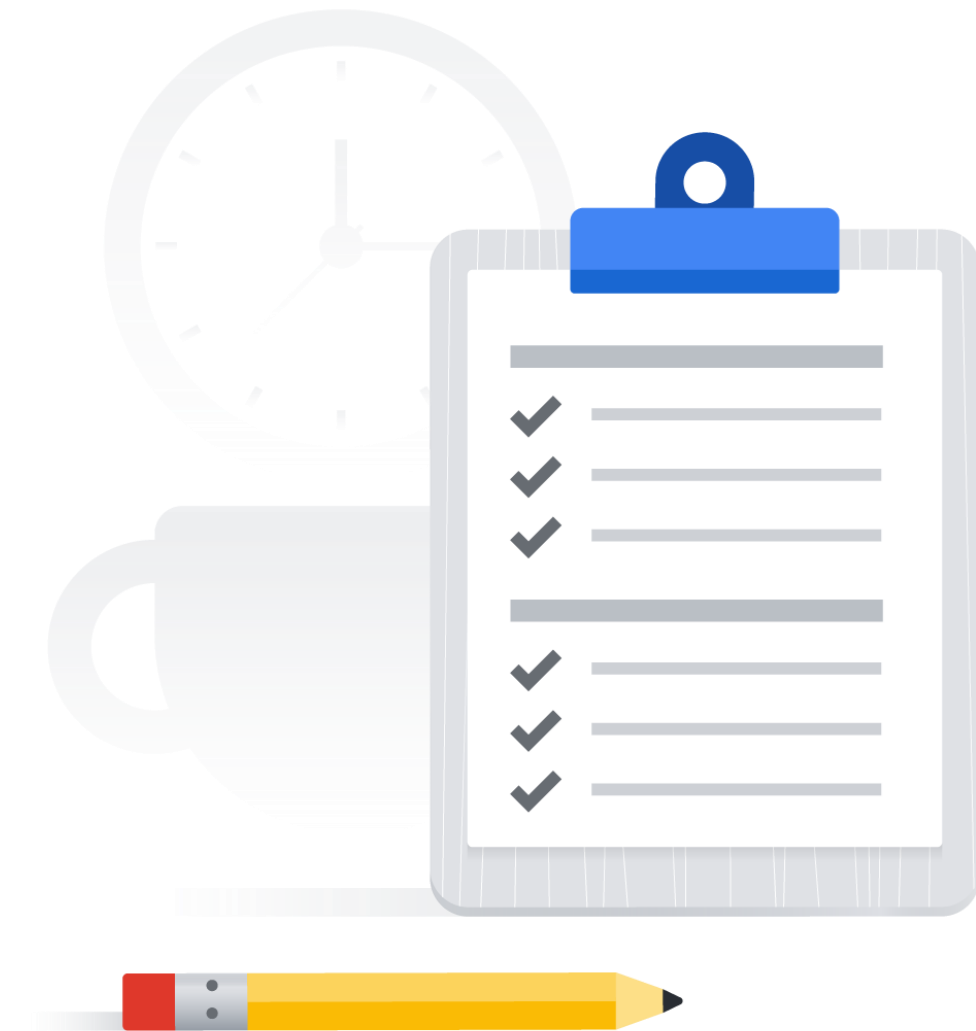
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Lab: Dataprep: Qwik Start

 Hands-on lab

During this lab you'll:

1. Use Dataprep to add a dataset.
2. Create recipes to interpret data into meaningful results.



- | | |
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Knowledge check

Which of Google Cloud's big data managed services is optimized for large-scale batch processing or long-running stream processing of structured and unstructured data?

- A. Dataproc
- B. Dataflow
- C. BigQuery
- D. Pub/Sub



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C. BigQuery

D. Pub/Sub



Knowledge check

Which of these is a managed Spark and Hadoop service that lets you benefit from open source data tools for batch processing, querying, streaming, and machine learning?

- A. Dataproc
- B. Dataflow
- C. BigQuery
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Knowledge check

Which of these is *not* a feature of BigQuery?

- A. BigQuery offers both data storage and analytics.
- B. BigQuery runs on your on-premises server.
- C. BigQuery offers a flexible pay-as-you-go pricing model.
- D. BigQuery provides built-in machine learning features.



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Knowledge check

You can use three basic patterns to load data into BigQuery. Which one involves using SQL statements to insert rows into an existing table or to write the results of a query to a table?

- A. Batch load
- B. Streaming
- C. Generated data



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Summary

- ✓ Explored big data managed services in the cloud.
- ✓ Examined using Dataproc to run Hadoop, Spark, and other big data technologies as a managed service in the cloud.
- ✓ Learned about building ETL pipelines as a managed service by using Dataflow.
- ✓ Explored BigQuery, a managed data warehouse and analytics engine.



